

BHARGAV VELAGA

Email: ybhargav751@gmail.com | Mobile: +1 (214)-799-1124 | [LinkedIn](#) | [Github](#)

SUMMARY

Experienced Data Engineer with 5+ years of experience specializing in data analysis, visualization, and pipeline optimization. Improved data accuracy by 20% through comprehensive exploratory data analysis and SQL-based data cleansing. Capable of creating actionable insights using Tableau and streamlining ETL processes. Proficient in Python, SQL, and AWS technologies, with demonstrated success in enhancing data processing efficiency and accelerating insights generation. Mastered Artificial Intelligence(AI), machine learning, and deep learning algorithms—such as Random Forest, XGBoost, Support Vector Machines (SVM), k-Means, and Neural Networks—to drive advanced analytics initiatives, streamlining data processing by 40% while ensuring compliance with industry regulations through robust pipeline maintenance.

EDUCATION

Master of Science in Information Technology
University of the Cumberland, Williamsburg.

May 2023 – August 2024

TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced), SQL (complex queries), Java, C#, R, Scala, Unix/Linux.
- **Data Engineering:** Apache Spark (Advanced), PySpark, Kafka, Flink, Hive, Hadoop, Spark SQL, Delta Lake
- **ETL & Orchestration:** AWS Glue, Airflow, Azure Data Factory, DBT, Fivetran, Informatica.
- **Cloud Platforms:** AWS (S3, Glue, Lambda, Redshift, EMR, Kinesis, Athena), Azure (Data Factory, Synapse, Data Lake), GCP
- **Databases:** Snowflake, Redshift, PostgreSQL, MySQL, MongoDB, Oracle.
- **BI Tools:** Power BI, Tableau, Looker, QuickSight, Excel.
- **DevOps & CI/CD:** Git, GitHub, Jenkins, Docker, Kubernetes, Terraform, CloudFormation.
- **Governance & Security:** IAM, Collibra, GDPR, HIPAA, SOC 2, CCPA.
- **ML & Tools:** SageMaker, TensorFlow, PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn.
- **Soft Skills:** Problem-Solving, Communication, Team Collaboration, Analytical Thinking, Attention to Detail.

PROFESSIONAL EXPERIENCE

Mayo Clinic
Data Engineer

August 2023 - Present
Rochester

Technologies: SQL, Python, Pandas, Tableau, NumPy, Matplotlib, MySQL, SQL Server, Power BI, Apache Cassandra, AWS(Glue, S3, Lambda, Redshift, Codepipeline), Apache Airflow, Teradata, DBT,Kubernetes, MongoDB, Jira.

- Developed real-time data ingestion pipelines with Python, Apache Kafka (MSK), and Kinesis Firehose to accelerate the processing of electronic health records (EHR), insurance claims, and IoT-generated patient data.
- Engineered and maintained scalable ETL/ELT pipelines using AWS Glue, AWS Lambda, and EMR, accelerating data processing speed by 40% and ensuring seamless data ingestion into Amazon Redshift and S3, supporting critical healthcare data workflows.
- Utilized DBT and AWS Data Pipeline to orchestrate data workflows and perform reliable transformations across structured and semi-structured datasets.
- Leveraged Amazon S3, Redshift, RDS (PostgreSQL), DynamoDB, and Amazon Keyspaces (Cassandra) to optimize data storage and retrieval, improving performance by 50% and reducing operational costs by 15% while ensuring HIPAA compliance.
- Implemented AWS S3 as a centralized data lake, integrating Databricks Delta Lake to support ACID transactions, versioned data, and faster analytical queries.
- Enhanced machine learning model performance by leveraging healthcare data APIs such as FHIR and HL7 to enrich patient datasets, leading to a 5% improvement in patient readmission prediction accuracy.
- Deployed Machine learning models in production environments using Docker, Kubernetes (EKS), and AWS SageMaker, reducing deployment inconsistencies by 35%.
- Implemented CI/CD pipelines with AWS CodePipeline and CodeBuild, automating model deployment, testing, and monitoring, reducing deployment time by 30%.
- Monitored production pipelines with AWS CloudWatch and AWS X-Ray, enabling proactive debugging and ensuring reliability and scalability of workflows.
- Automated encryption key rotation and access auditing using AWS CloudTrail and AWS KMS, ensuring compliance with HIPAA regulations and enhancing data security by reducing the risk of unauthorized access to protected health information (PHI)
- Automated data reporting and visualization using Pandas, NumPy, Matplotlib, and Seaborn, reducing manual reporting by 40% and enabling actionable insights.
- Created interactive dashboards in Amazon QuickSight and Tableau, boosting stakeholder engagement and supporting data-driven decision-making by 25%.

Jenius Bank

Data Engineer

August 2022 – July 2023

Hartford, CT

Technologies: AWS (Lambda, Glue, Data Pipeline), DBT, Snowflake, Teradata, Tableau, Lambda, SQL, Python, ETL, Docker, GitHub, Pandas, Numpy, Terraform.

- Devised and formulated end-to-end data pipelines utilizing AWS Lambda, Step Functions, Data Pipeline, and Databricks, orchestrating complex workflows and ensuring efficient, reliable data movement while reducing processing latency and ingestion time by 35%.
- Applied Snowflake features like Snow Pipes and stages to process 100GB data every day in real-time and ingest the data into Snowflake tables.
- Established ETL processes using AWS Glue, Step Functions, and Python scripts to extract, transform, and load data from multiple sources into Snowflake, streamlining data integration workflows by 35%.
- Harnessed Snowflake's cloud-native data warehousing capabilities to build and manage scalable, high-performance data storage solutions, improving query execution speed by 50%.
- Created SQL Server stored procedures, functions, views, and triggers using both SQL and T-SQL with an improvement of 30% in performance.
- Optimized data pipelines for ERP data processing using AWS Lambda and Databricks, reducing cloud costs by 30% while maintaining high data throughput.
- Enhanced big data processing using AWS Databricks, implementing PySpark and MLflow for scalable ETL workflows and machine learning model tracking, reducing processing time by 40%.
- Performed numerical data transformations and statistical analysis with NumPy and Pandas, reducing computational time by 35% and improving insight generation by 25%.
- Implemented data quality checks, validations, and monitoring to ensure data integrity, consistency, and reliability, reducing discrepancies by 40%.
- Optimized Tableau dashboards by implementing efficient calculations and data extracts, leveraging LOD expressions, filters, and parameter controls to reduce query execution time by 35% and improve dashboard responsiveness.

IBM

Data Engineer

May 2021 – July 2022

India

Technologies: AWS (S3, Redshift, Lambda, EC2), IBM DataStage, SAS, SQL, Python, PySpark, RDS, CloudWatch, Terraform (basic), Linux, Git, JIRA, Oracle, and MS SQL Server.

- Supported ETL workflows and data migration processes using AWS services like S3, Redshift, and Lambda.
- Assisted in configuring AWS environments and performing basic monitoring and maintenance tasks.
- Developed SQL scripts and Python routines for data validation, transformation, and report preparation.
- Worked with DataStage and SAS for integrating and transforming data from on-prem and cloud sources.
- Collaborated with senior engineers to implement best practices for data processing and AWS resource management.

CERTIFICATIONS

- **AWS Certified Associate Data Engineer: Amazon**